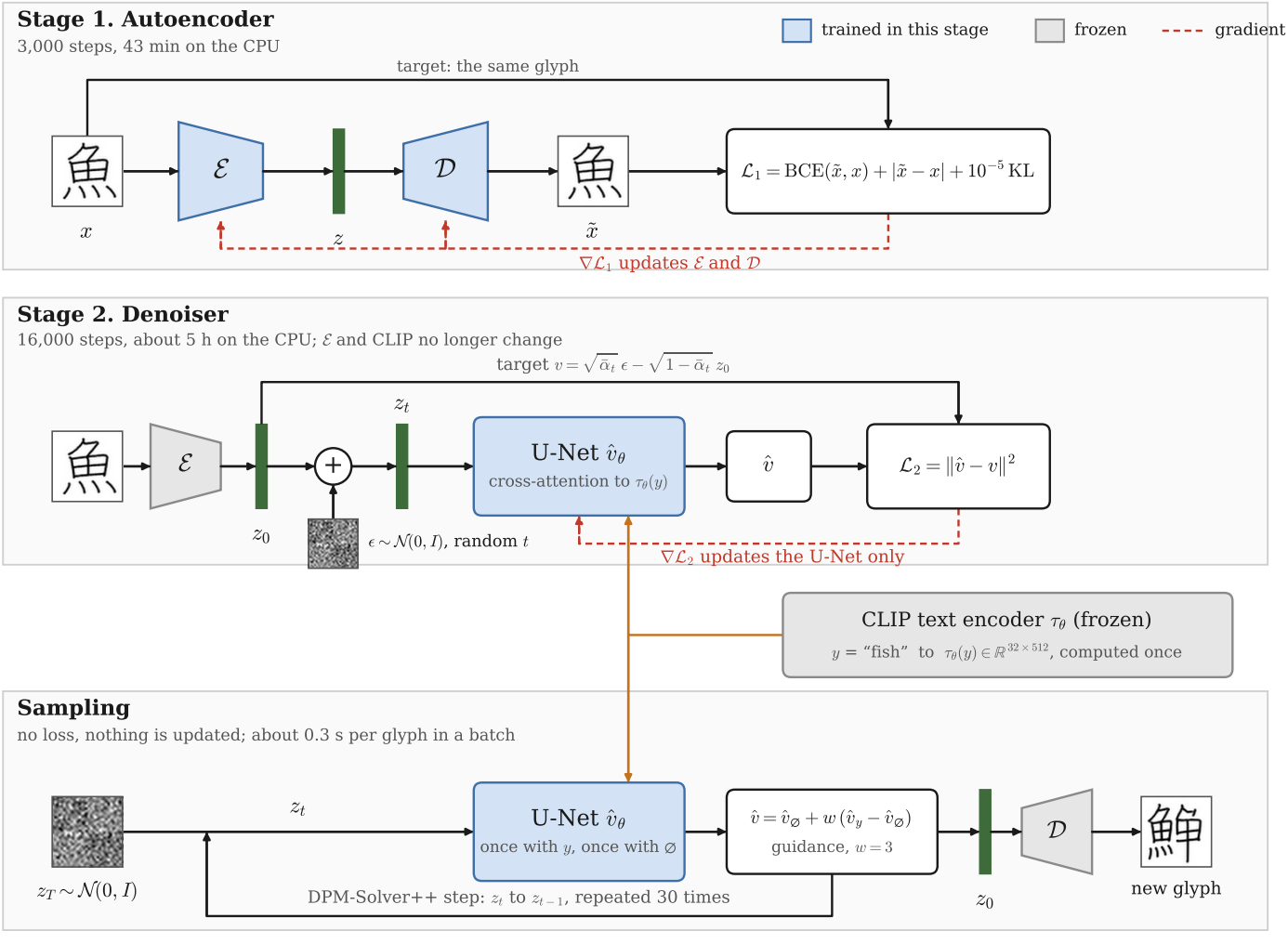

Stage 1. Autoencoder
3,000 steps, 43 min on the CPU
trained in this stage
frozen
gradient
target: the same glyph
$\mathcal{L}_1 = \mathrm{BCE}(\tilde{x}, x) + |\tilde{x} - x| + 10^{-5}\,\mathrm{KL}$
x
z
$\tilde{x}$
$\nabla\mathcal{L}_1$ updates $\mathcal{E}$ and $\mathcal{D}$
Stage 2. Denoiser
16,000 steps, about 5 h on the CPU; $\mathcal{E}$ and CLIP no longer change
target $v = \sqrt{\bar{\alpha}_t}\,\epsilon - \sqrt{1-\bar{\alpha}_t}\,z_0$
z_t
U-Net $\hat{v}_\theta$
cross-attention to $\tau_\theta(y)$
$\hat{v}$
$\mathcal{L}_2 = \|\hat{v} - v\|^2$
z_0
$\epsilon \sim \mathcal{N}(0, I)$, random t
$\nabla\mathcal{L}_2$ updates the U-Net only
CLIP text encoder τ_θ (frozen)
y = "fish" to $\tau_\theta(y) \in \mathbb{R}^{32\times 512}$, computed once
Sampling
no loss, nothing is updated; about 0.3 s per glyph in a batch
z_t
U-Net $\hat{v}_\theta$
once with y, once with $\varnothing$
$\hat{v} = \hat{v}_\varnothing + w\,(\hat{v}_y - \hat{v}_\varnothing)$
guidance, $w = 3$
z_0
new glyph
$z_T \sim \mathcal{N}(0, I)$
DPM-Solver++ step: z_t to z_{t-1}, repeated 30 times